Sahaj Bhadja

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Summary

Backend Software Engineer with 2+ years building **Java** and Python services, RESTful APIs on **Postgres**, time-series data pipelines, and event-driven integrations on **AWS**, with testing, code review, monitoring, and production troubleshooting.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized **Postgres** schemas with indexing and query tuning for 200+ users and 10K+ records, cutting latency 35%, and cut dashboard latency from 1.2s to 350ms for 500+ users.
- Built fault-tolerant background services with explicit job states and failure isolation, tripling throughput across 1K+ jobs, and **monitored** 5K+ monthly production jobs at 99%+ reliability through CI/CD releases.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built **RESTful APIs** and backend services on AWS with MySQL on RDS for upload, rubric, and grading workflows, serving 150+ instructors and 10K+ daily requests from a React frontend.
- **Troubleshot** production timeouts and model failures with fallback routing and multithreaded batch processing, keeping services stable at peak load and cutting instructor feedback turnaround 60% across 10K+ submissions.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built Python pipelines processing 100K+ **time-series** trajectory logs sampled at 10Hz into aligned features, training a PyTorch planner that improved long-horizon trajectory planning 18% in closed-loop simulation.
- Benchmarked a new spatial-temporal attention encoder fusing map and agent data against three baselines, documenting failure modes and **insights** while improving closed-loop planning stability 15% in simulated driving.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built object-oriented **Java** and Python microservice modules with Singleton and Factory patterns, covered by 20+ Mockito and PyTest suites in nightly Jenkins pipelines, and cut technical debt 20%.
- Redesigned **AWS** Glue and S3 pipelines that consolidated user activity logs from multiple sources, boosting efficiency 68%, and optimized APIs with caching to improve response times 30%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **event-driven** payment processing with idempotent webhooks for refunds, disputes, and status updates, cutting manual handling about 70% and keeping payment status consistent across 50K+ booking records.
- Built Spring Boot and Flask APIs for bookings and pricing with Firebase real-time sync to Android, iOS, and web clients, keeping responses under 200ms and confirmations 40% faster.

Skills

Languages: Java, Python, SQL, JavaScript, TypeScript, Debugging, Firebase, MongoDB, Selenium
Backend: Spring Boot, FastAPI, Flask, REST APIs, Microservices, Webhooks, Event-Driven Systems
Data: PostgreSQL, MySQL, SQL Server, Schema Design, Query Optimization, Spark, Time-Series Data
Cloud & Ops: AWS (S3, RDS, Glue, EC2), Docker, Kubernetes, CI/CD, Jenkins, Sentry, Monitoring
Practices: Data Structures, Algorithms, System Design, Unit Testing, Mockito, PyTest, Code Reviews, Git, Agile, Linux

Projects

Wearable Sensor Ingestion Service | *Java, Spring Boot, Kafka, Postgres, AWS*

- Designed a **Spring Boot** service that ingests heart-rate and motion samples from wearables through **Kafka**, deduplicates late or out-of-order batches, and stores per-member time-series data in Postgres.

Daily Recovery Score API | *Java, Postgres, Kafka Streams, REST APIs, JUnit*

- Designed a **REST API** that computes a daily recovery score from overnight heart-rate variability and sleep data, caches results per member, and exposes trends for the app with **JUnit** tests.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, Docker, Maven*

- Extended a **Java** and TypeScript database workspace supporting 10+ engines including PostgreSQL, with schema synchronization and AI-generated SQL reaching 82% accuracy, built with Maven and Docker across Windows, macOS, and Linux.